

Introduction to the Machine Learning Course

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Profile



張詠淳 (Yung-Chun Chang)

http://nlp.tmu.edu.tw/

- **Work Experience**
  - Director, TMU, Graduate Institute of Data Science, (2025/3–present)
  - Professor, TMU, Graduate Institute of Data Science, (2023/8–present)
  - Deputy Chief Data Officer, TMU, Office of Data Science, (2020/8–2025/02)
  - Associate Professor, TMU, Graduate Institute of Data Science, (2020/8–2023/7)
  - Assistant Professor, TMU, Graduate Institute of Data Science, (2017/8–2020/7)
  - Postdoctoral Research Fellow, Academia Sinica, Institute of Information Science, (2016/8–2017/7)
  - Research Assistant, Academia Sinica, Institute of Information Science, (2008/1–2016/7)
- **Research Interests**
  - Natural Language Processing
  - Information Extraction and Retrieval
  - Business Intelligence
  - Biomedical Text Mining
  - AI in EHRs

You will learn



- Basic concept of machine learning, and deep learning.
- Basic technology of data science (e.g. preprocessing, data representation, performance evaluation…)
- How to develop machine learning and deep learning models.
- You will learn how to develop a machine learning and deep learning model for the applications of NLP and CV.

Course Information

- Instructor:
  - 張詠淳 (Yung-Chun Chang)
  - E-mail: changyc@tmu.edu.tw
  - Office hour: after class of by appointment
- Class time: 15:10~17:10, every Wednesday.
- Classroom: 9703 ShuangHo Campus
- Teaching assistants:
  - 葉文照 (wcyeh@proton.me)
- Course website:
  - IM@TMU(https://im.tmu.edu.tw/course/19514)





Why use Python for AI?

The basic procedure of developing an ML model

- **The Data**: Data collection, Data visualization, Data preprocess, Data representation….
- **Model Development**: Machine learning and deep learning, performance evaluation….
- **Production and Engineering**: web-based service, GUI, Unit-test…

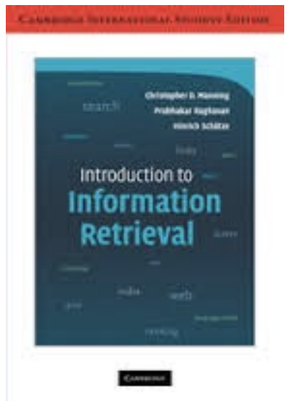


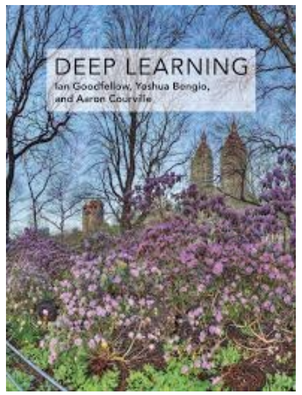
Grading Policy

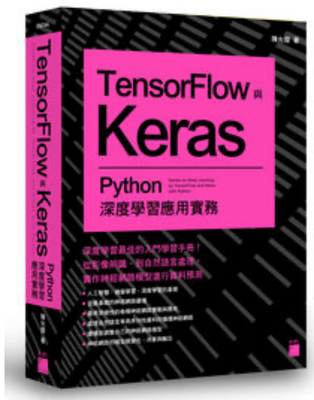
- Midterm Exam: 30% [Week 9 (11/04)]
- Kaggle Competition: 30% [Week 6 – 14]
- Final project: 40% [Week 16 (12/23)]

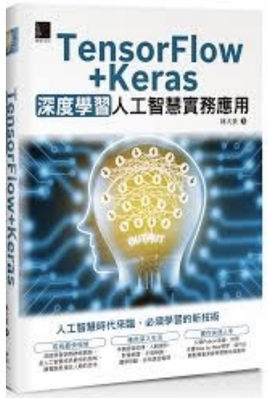
Textbook

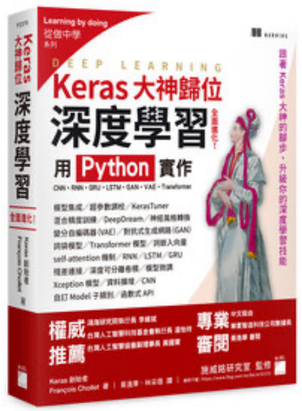
- Christopher D. Manning, Prabhakar Raghavan, and Hinrich Sch¨utze, Introduction to Information Retrieval, Cambridge University Press, 1st ed., 2008. (https://nlp.stanford.edu/IR-book/information-retrieval-book.html)
- Ian Goodfellow, Yoshua Bengio and Aaron Courville, Deep Learning, MIT Press, 1st ed., 2016.
- 林大貴, TensorFlow Keras深度學習人工智慧實務應用, 博碩, 2018
- Keras 大神歸位：深度學習全面進化！用 Python 實作CNN、RNN、GRU、LSTM、GAN、VAE、Transformer, 旗標科技
- tf.keras 技術者們必讀！深度學習攻略手冊, 旗標科技
- TensorFlow 與 Keras - Python 深度學習應用實務, 旗標科技

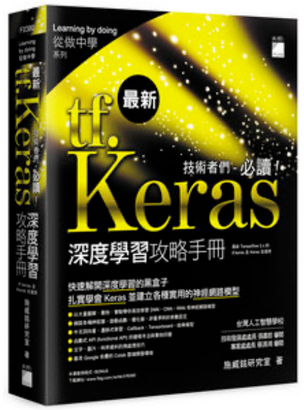




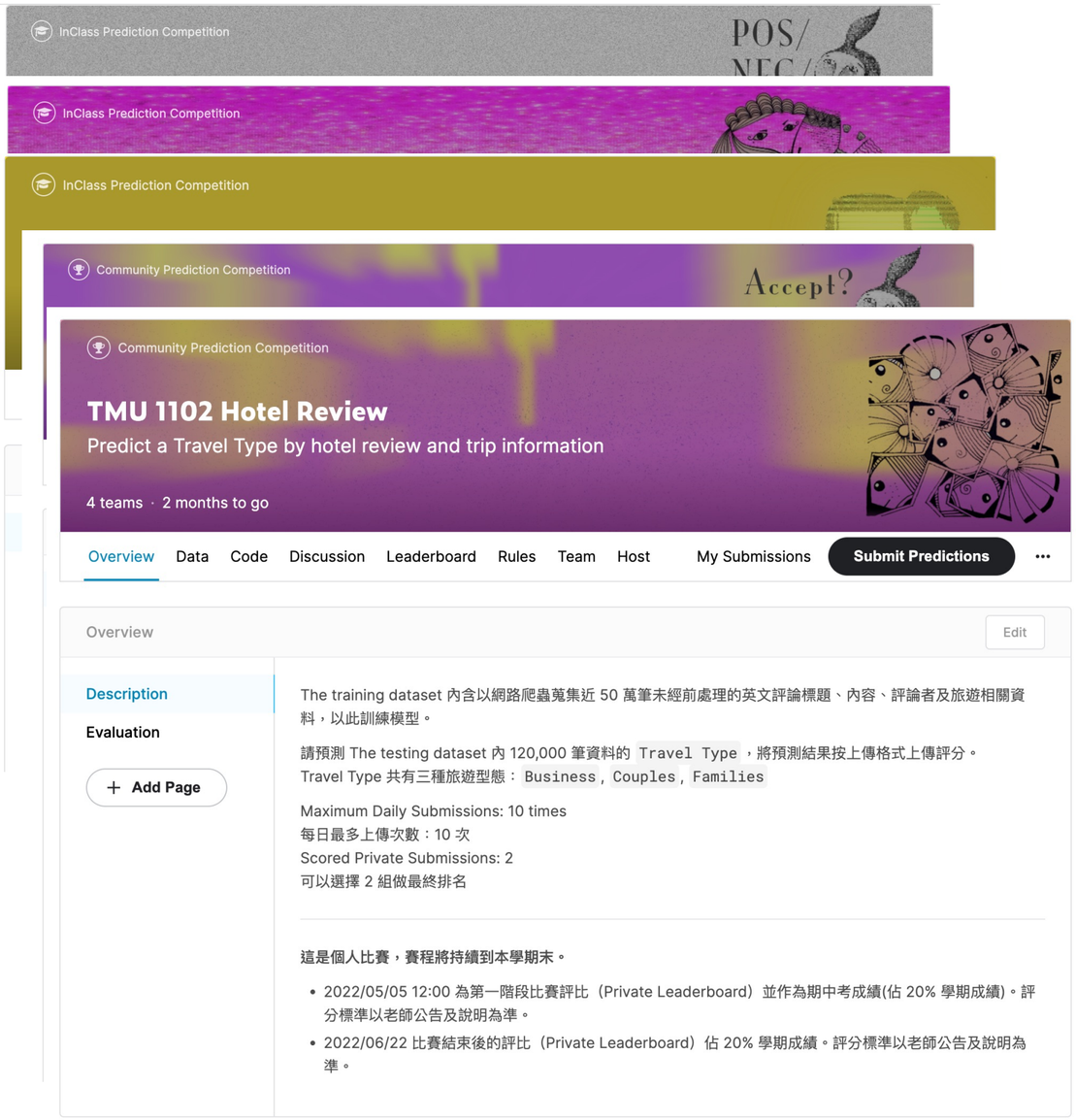






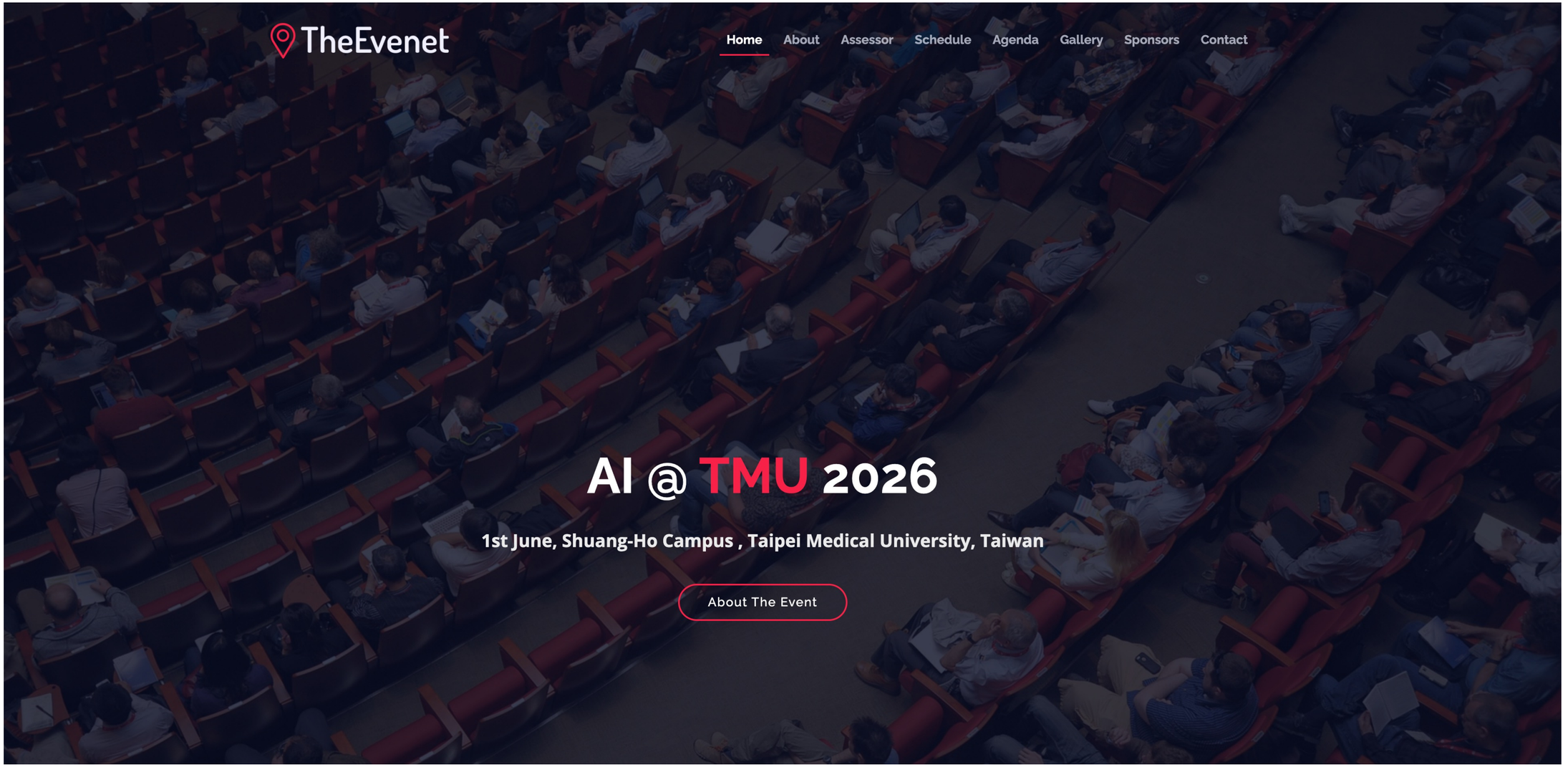


Kaggle Competition



#	Team	Members	Score	Entries	Last	Code
	Baseline:80分		0.67044			
1	助教測試		0.67044	16	18h	
	Baseline:70分		0.63477			
	Baseline:70分		0.63477			
2	昆圖庫塔卡提考特蘇瓦西拉松		0.59569	1	3d	
	Baseline:0分		0.59127			
3	adama20		0.59127	2	5d	
4	TOOTHACHE		0.50933	7	21h	
4	—	還沒想到Team Name	0.86638	10	14d	
5	—	排球隊	0.86327	31	12d	
6	—	跳脫公衛二人組	0.85612	36	18d	
7	—	白爛貓	0.85325	11	13d	
8	—	DL Team	0.85325	1	14d	
9	—	我是一隻樹懶	0.85325	3	13d	
10	▲1	美少女戰士	0.84418	11	13d	
11	▼1	魔王拍張照	0.84323	28	12d	
12	—	軒軒毛毛gogo	0.77785	2	14d	
13	—	十二萬分的感謝隊	0.71581	2	19d	
14	—	TMU o902108011	0.70150	3	17d	
15	—	可能沒有這一隊	0.52469	4	25d	
		烏白臆（每組都要比這筆好）	0.49176			

AI@TMU Competition



https://nlp.tmu.edu.tw/AITMU/

Final Project Guidelines

- **Topic**: No restriction on the topic, but it must be related to Machine Learning.
- **Requirements**: Your project should include:
  - A clear data source (can be publicly available or self-collected)
  - Data analysis and preprocessing
  - Model training
  - Performance evaluation
- **Team Formation**: 2–3 students per group.
- **Deliverables**:
  - Demo Video showcasing your project
  - Final Project Presentation: 10 minutes + 2 minutes Q&A





Thanks for your listening

Q & A